Further Strategies — problem 14.32

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(a) For every $n \in \mathbb{N}$, we have $n\sqrt{2} - \lfloor n\sqrt{2} \rfloor > 1/(2n\sqrt{2})$.
(b) For every $\epsilon > 0$ there is an $n \in \mathbb{N}$ such that $n\sqrt{2} - \lfloor n\sqrt{2} \rfloor < (1+\epsilon)/(2n\sqrt{2})$.